This model was widely accepted because of the following features

1. The densely packed bimolecular lipid layer shows the presence of 40% lipid by weight in the membrane.
2. The model accounted for the three layered staining pattern of fixed membranes as observed in electron microscopy.
3. Phospholipids spontaneously form a bimolecular system in vitro when added to an aqueous environment and there is no requirement for work input to maintain the minimum energy conformation of the synthetic membrane.
4. The membrane is rich in hydrocarbons, so it is of high electrical resistance.
5. High permeability of natural membranes to non-polar molecules could be explained by their solubility in the non-polar lipid phase and at the same time accounted for relative impermeability to small ions which do not dissolve readily in this medium.

Drawbacks

(i) The thickness of the membrane was found to be greater in plasma membrane than in the intracellular membrane of endoplasmic reticulum or golgi complex

(ii) This model is based on the study of the myelin sheath of a nerve fibre, which is a non-typical membrane and hence cannot be used as a reference model for the cell membrane

1.2.5.Fluid Mosaic Model:

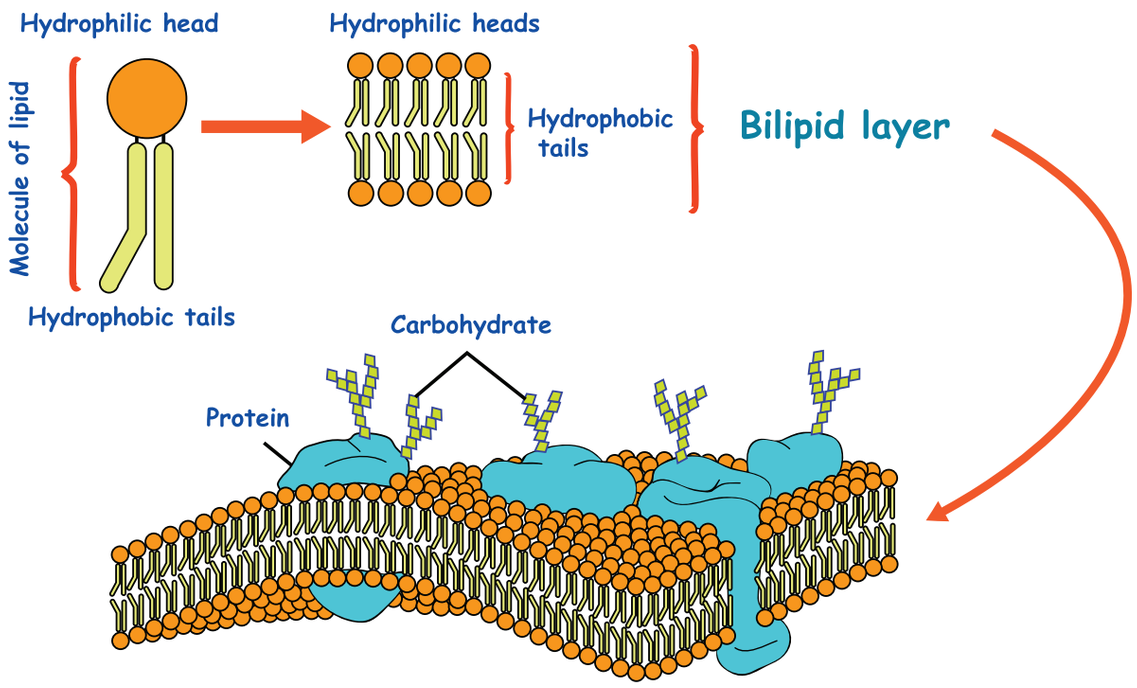


Figure 1.5: Fluid membrane bilayer

The fluid mosaic model of cell membrane was proposed in 1972 by S.J. Singer and G.L. Nicolson. The plasma membrane is a unique component of both plant and animal cells. It serves as a barrier between the cell interior and its surrounding and it allows entry

XII U1 Cell Membrane.indd 8 4/8/2020 3:03:29 PM